



Beginner Cybersecurity Roadmap

A comprehensive, structured path from zero to cyber professional — follow each phase sequentially for best results

📖 Phase 1 · Begin

⚙️ Phase 2 · Interim Skill

🔍 Phase 3 · Continuous Learning

3
Phases

4
Certifications

20+
Skills



Phase 1 — Begin

Start here · Build your foundation

SELF-ASSESSMENT

- Identify Strengths & Weaknesses**
Evaluate your current knowledge of networking, OS concepts, and problem-solving — be honest about gaps
- Set Clear, Measurable Goals**
Define your target role — SOC analyst, pentester, cloud security engineer — and set a realistic 6–12 month timeline

BASIC TRAINING — CORE KNOWLEDGE AREAS

- 🌐 Networking Fundamentals
- 💻 Operating Systems
- 🔒 Basic Security Concepts
- 🔑 Introduction to Cryptography
- 🌐 Web Security Basics
- 🚒 Incident Response & Handling
- 🔍 Vulnerability Assessment & Pentesting
- 📡 Network Security Protocols & Technologies

💡 Pro Tip — Foundation First

Master TCP/IP, the OSI model, and the Linux command line before moving to Phase 2. These are the bedrock of every cybersecurity discipline. Skipping this step will slow you down later.

RECOMMENDED STARTING RESOURCES

- 🎮 **CompTIA IT Fundamentals (ITF+)**
Great first certification — covers hardware, software, networking & security basics
- 🎮 **Professor Messer's Free Study Guides**
Excellent free video series covering Security+ and Network+ domains
- 🎮 **Linux Journey (linuxjourney.com)**
Interactive, free Linux fundamentals training — essential for any security path



Phase 2 — Interim Skill

Specialize · Practice · Certify

SPECIALIZATIONS — CHOOSE YOUR FOCUS PATH

- 🌐 Network Security
- 💻 Application Security
- ☁️ Cloud Security
- 🔍 Incident Response & Forensics
- 🔍 Threat Intelligence Analysis
- 💻 SOC Management
- 🔍 VAPT — Vulnerability Assessment & Penetration Testing
- 💻 Secure Coding Practices

HANDS-ON EXPERIENCE — GET YOUR HANDS DIRTY

🔧 Labs & Practice Environments

- TryHackMe** — Beginner-friendly guided rooms & structured learning paths
- HackTheBox** — Intermediate to advanced realistic machine challenges
- DVWA** — Deliberately Vulnerable Web App for safe web security practice

🚩 Capture The Flag (CTF) Competitions

- CTFtime.org** — Global CTF event calendar & team finder platform
- PicoCTF** — Carnegie Mellon's beginner-friendly annual competition
- Regular CTF practice sharpens real-world problem-solving under pressure

KEY CERTIFICATIONS — VALIDATE YOUR SKILLS

CompTIA Security+

SY0-701 · Globally recognized entry-level security credential · Vendor-neutral · Exam fee ~\$392

CEH — Certified Ethical Hacker

EC-Council · Covers 20 ethical hacking domains · Ideal for pentesters & red teamers

OSCP — Offensive Security Certified Professional

24-hour hands-on exam · Gold standard for penetration testing professionals worldwide

CCNA Cyber Ops

Cisco · Network security operations · Ideal for SOC analyst career track



Phase 3 — Continuous Learning

Stay sharp · Engage · Give back

STAY UPDATED — NEVER STOP LEARNING

- 🗣️ **Conferences & Workshops** — Attend DEF CON, Black Hat, BSides events to hear cutting-edge research
- 🎓 **Online Courses & Tutorials** — Coursera, Udemy, SANS Institute, Cybrary, INE Security
- 🗣️ **Networking Events & Meetups** — Local ISACA, (ISC)² and OWASP chapter meetings
- 📺 **Webinars & Virtual Conferences** — Free live sessions from Palo Alto, CrowdStrike, SANS
- 💬 **Communities & Forums** — Reddit r/netsec, Discord servers, OWASP Slack, Hack Forums
- 📰 **Blogs & Newsletters** — Krebs on Security, Schneier on Security, SANS Internet Stormcast
- 👤 **Follow Industry Experts** — Twitter/LinkedIn: researchers, CISOs, threat hunters, bug bounty hunters
- 📖 **Books & Whitepapers** — NIST guidelines, CVE advisories, vendor threat intelligence reports
- 🚩 **CTF Competitions** — Compete regularly on CTFtime to benchmark and grow your skills
- 💰 **Bug Bounty Programs** — HackerOne and Bugcrowd: earn real rewards while building your portfolio

MENTORSHIP — GIVE BACK & GROW TOGETHER

🌱 Mentor Others

Teaching newcomers reinforces your own knowledge and develops leadership skills. Share write-ups, record tutorials, or volunteer at local cybersecurity workshops and schools.

🔍 Seek Guidance from Experts

Connect with experienced professionals via LinkedIn, local chapter events, or platforms like MentorCruise. A strong mentor can accelerate your career by years and help you avoid costly mistakes.